

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2023/2024

JULY EXAMINATION

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

WEDNESDAY, 10 JULY 2024

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

- a) IOI City Mall has recently introduced an automated *License Plate Recognition (LPR)* system for its car parking at all levels. The system allows shoppers to drive through upon entry and exit the parking without the issuance of printed parking tickets as it will capture vehicle plate numbers using the *LPR* camera installed at the parking entry and exit points.
- (i) *PEAS*, which stands for *Performance, Environment, Actuator* and *Sensor*, is an effective way of describing how an intelligent system works. Construct a *PEAS* description for the problem to which the *LPR* is going to solve. (4 marks)
 - (ii) Based on information provided below, identify **SIX (6)** properties of *LPR*'s environment. Justify your answer. (6 + 6 marks)
 - Fully observable, partially observable or non-observable
 - Episodic or sequential
 - Single-agent or multi-agent
 - Static or dynamic
 - Deterministic or non-deterministic
 - Discrete or continuous (in terms of percept)
- b) *ChatGPT* is a different type of intelligent system from the *LPR*. It is an *Artificial Intelligence (AI)* chat-bot that can create human-like conversational dialogue. Two months right after its first launch at the end of year 2022, it already has more than 100 million active users. Its achievement has surpassed many users' expectations. One important reason for its popularity is its learning capability.
- (i) A learning agent usually consists of four *conceptual components*. Identify the **FOUR (4)** *conceptual components*. Then, briefly describe each of them. (4 + 4 marks)
 - (ii) Name **ONE (1)** person who is credited to the method of creating intelligent machines, that is, first build learning machines and then teach them later. (1 mark)

[Total: 25 marks]

Question 2

- a) *Adversarial search* is one important type of search in *AI*. *Adversarial search problems*, also known as *games*, normally occur in multi-agent competitive environments.
- (i) Explain why the term "*adversarial search*" is named so. (2 marks)
 - (ii) Differentiate *multi-agent environment* from *single-agent environment*. (2 marks)
 - (iii) Differentiate *competitive environment* from *cooperative environment*. (2 marks)
 - (iv) Identify **FIVE (5)** common characteristics of *games* that are most studied in the field of *AI*. (5 marks)

Question 2 (Continued)

- b) A game tree is a kind of search tree consisting of nodes and edges that represent all possible legal moves of a game, where nodes are positions, and the edges are moves in a game. Answer the following questions based on the game tree provided in **Figure 1** below:

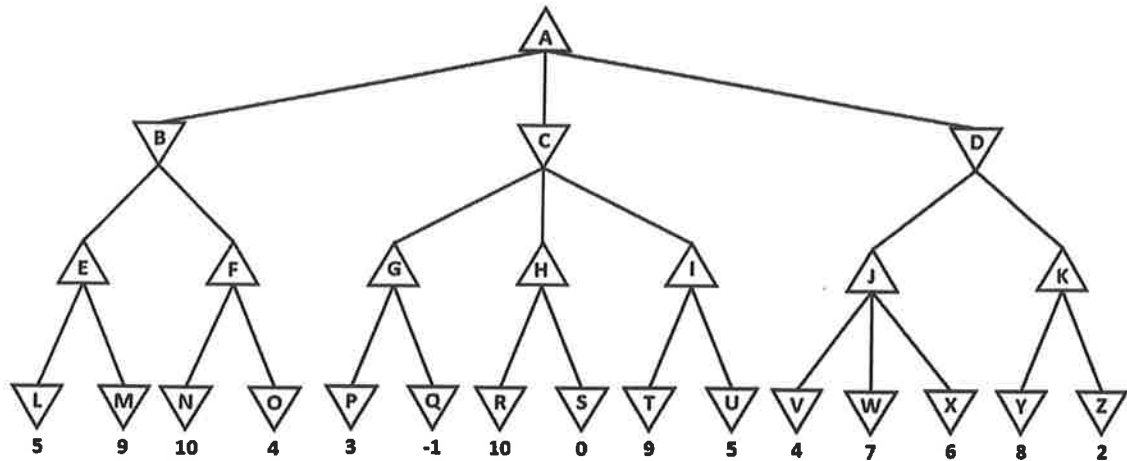


Figure 1. Game tree

- (i) Suppose the *minimax* algorithm is used to explore the game tree in search of an optimal move, compute the *minimax* value for nodes B, C, D, E, F, G, H, I, J, and K. (5 marks)
- (ii) Identify the best move for the *MAX* player at node A. Justify your answer. (1 + 1 marks)
- (iii) Briefly describe what will happen when *alpha-beta pruning* is applied to a *minimax* tree. (2 marks)
- (iv) Suppose *minimax* search is applied to the game tree with *alpha-beta pruning* performed in left-to-right order, deduce nodes that would be pruned. (5 marks)

[Total: 25 marks]

Question 3

- a) Suppose $P_i = \{P_1, P_2, P_3, P_4\}$ represents four participants of a *Creative Content Competition* who are nominated for the *Best Creative Idea Award*. The domain of each variable is the set $D_i = \{0, 1\}$ where 0 and 1 mean “not awarded” and “awarded” respectively. For instance, when $P_i = 1$, it means that P_i will receive the award; and when $P_i = 0$, it means that P_i will not receive the award. For each of the following statement, write a constraint in **TWO (2)** different ways: 1) an *abstract relation*, and 2) an *enumeration of the members of the relation*; and then indicate whether it is a *unary constraint*, a *binary constraint*, or a *ternary constraint*.
- (i) Only **ONE (1)** participant will be awarded. (3 marks)
 - (ii) At least **TWO (2)** participants between P_1, P_2 and P_3 will be awarded. (3 marks)
 - (iii) Only **TWO (2)** participants will be awarded. (3 marks)
 - (iv) P_2, P_3 and P_4 will be awarded or will not be awarded at the same time. (3 marks)

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 3 (Continued)**

- b) Sudoku is a puzzle game based on the logical placement of numbers which has introduced millions of people to *Constraint Satisfaction Problem (CSP)*. **Figure 2** shows an example of 4×4 Sudoku consists of 16 squares, some of which are initially filled with digits between 1 and 4. The objective of solving a Sudoku is to fill the 4×4 grid with digits so that each column, each row, and each of the four 2×2 sub-grids that compose the grid contain digits between 1 and 4. Answer the following two questions based on the Sudoku provided in **Figure 2**.

	1	2	3	4
A		3	1	2
B	2			4
C	3			1
D			4	

Figure 2. Sudoku

- (i) Formulate the Sudoku as a *CSP* by identifying its variables, domains for each variable and constraints using the *Alldiff* global constraint. (5 marks)
- (ii) Suppose the *Alldiff* constraints have been expanded into binary constraints so that the *AC-3 algorithm* can be applied directly, explain how **TWO (2)** variables, *C3* and *B2*, are made arc-consistent step-by-step. It is important to note that all steps applied in making *C3* arc-consistent will remain effective when it is *B2*'s turn to be made arc-consistent. (4 + 4 marks)

[Total: 25 marks]

Question 4

- a) Knowledge-based (*KB*) agents are more powerful than problem-solving agents. With the use of logic as a general class of representation, they can combine and recombine information to draw a conclusion based on various conditions to suit a lot of purposes. This impressive capability can be realized through the implementation of two core components which are *knowledge-base* and *inference system*.
- (i) Briefly describe *knowledge-base* and *inference system*. (2 marks)
 - (ii) Create an illustration or drawing with the use of the following **SIX (6)** components to explain how a *KB* agent works. (2 marks)
 - Environment
 - Inference engine
 - Input
 - Knowledge base
 - Learning
 - Output

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4 (Continued)**

- b) In propositional logic, a proposition refers to a sentence which is either *True* or *False*, but it cannot be both. Usually, propositions are represented with the use of symbols. Referring to information given in **Table 1**, translate the following ten English sentences into propositions with the use of symbols and appropriate logical connectives (only if applicable).

Table 1: Propositions and their symbols

Symbol	Proposition
A	I like AACS3273.
B	I will pass AACS3273.
C	I will fail AACS3273
D	AACS3273 is tough.
E	AACS3273 is interesting.

- (i) I like AACS3273 and I will pass it. (1 mark)
- (ii) AACS3273 is not a tough subject. (1 mark)
- (iii) I will not fail AACS3273, but I will pass it. (1 mark)
- (iv) AACS3273 is interesting but I will not pass it. (1 mark)
- (v) It is not the case that if AACS3273 is interesting, then I will pass it. (1 mark)
- (vi) I will either pass AACS3273 or fail it. (1 mark)
- (vii) If AACS3273 is tough and I dislike it, then I will not pass it. (1 mark)
- (viii) I will pass AACS3273 if and only if it is interesting and I like it. (1 mark)
- (ix) If I dislike AACS3273, then it is not interesting. (1 mark)
- (x) It is not the case that, if AACS3273 is not tough and I like it, then I will either pass it or not failing it. (1 mark)

A *truth table* is a complete list of possible truth values of a given proposition. Construct a *truth table* for each of the following propositions and identify if it is a *tautology*.

- (xi) The proposition created as the answer to **Question 4 b) (ix)**. (4 marks)
- (xii) The proposition created as the answer to **Question 4 b) (x)**. (7 marks)

[Total: 25 marks]